

SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				$C_7H_{14}O_2$		1		1		
2.				award (1) for either of following $C_3H_8 + 3\frac{1}{2}O_2 \rightarrow 3CO + 4H_2O$ $2C_3H_8 + 7O_2 \rightarrow 6CO + 8H_2O$		1		1		
3.				hydrogen	1			1		
4.				award (1) each for any two of following $\begin{array}{cccc} H & H & H & H \\ & & & \\ H-C & -C & -C & =C \\ & & & \\ H & H & & H \end{array}$ $\begin{array}{cccc} H & H & H & H \\ & & & \\ H-C & -C & =C & -C-H \\ & & & \\ H & & & H \end{array}$ $\begin{array}{ccc} H & & H \\ & & \\ H-C & -C & =C \\ & & \\ H & C & H \\ & & \\ H-C & -H \\ \\ H \end{array}$ $\begin{array}{c} CH_3 \\ \\ CH \\ / \quad \backslash \\ H_2C \quad CH_2 \end{array}$ accept both <i>E</i> and <i>Z</i> isomers of but-2-ene accept skeletal formulae		2		2		

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					AO1	AO2	AO3	Total	Maths	Prac
5.				award (1) for two correct structures H H H H—C—C—C—H H H Br H H H H—C—C—C—H H Br H 2-bromopropane (1)		2		2		
6.				40 / +40 do not accept –40	1			1		
7.				award (2) for Z-3-bromohex-3-ene award (1) for either of following 3-bromohex-3-ene E-3-bromohex-3-ene no credit for any other answer		2		2		
				Section A Total	2	8	0	10	0	0